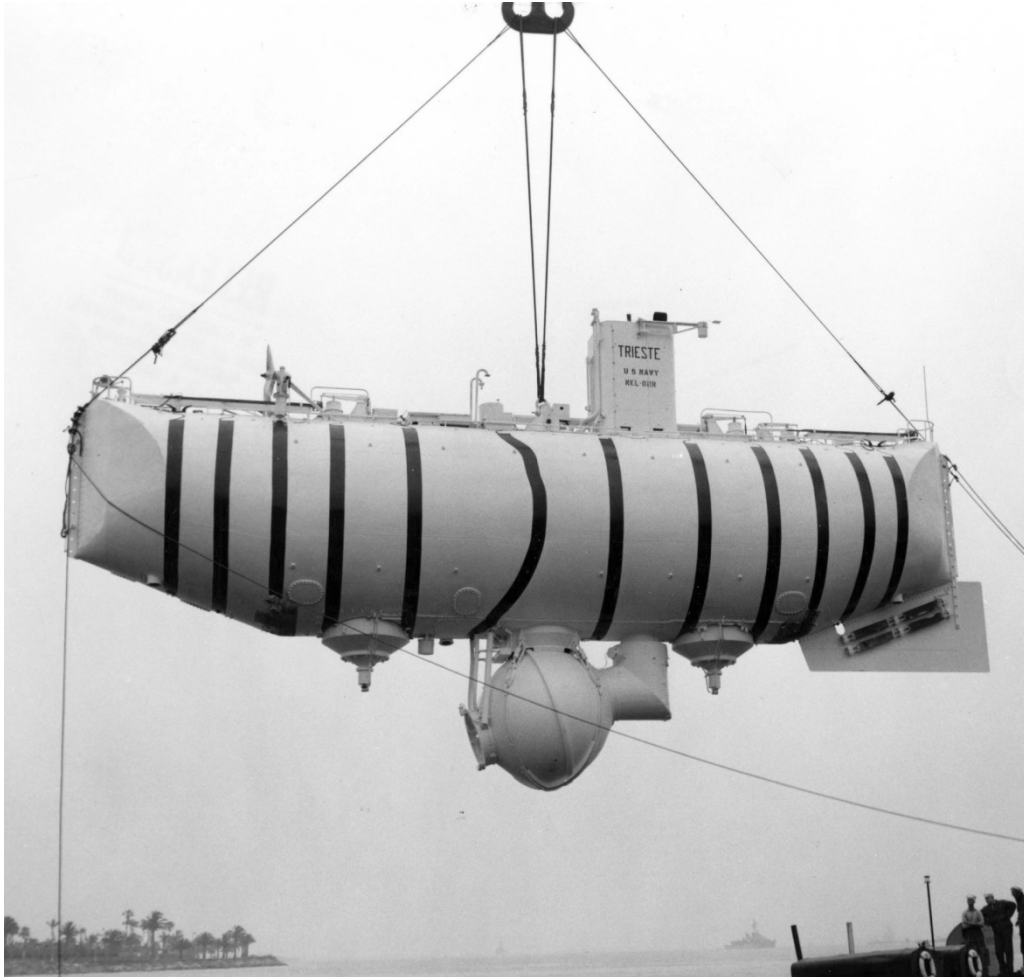


Chapter 2 Pressure Distribution in a Fluid

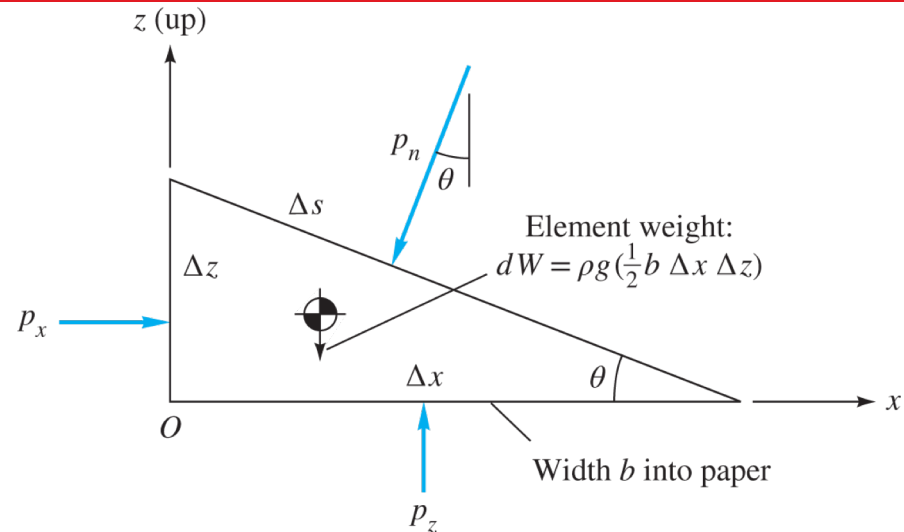


Learning Objectives

- Recognize pressure variation in a static fluid.
- Calculate pressure distribution in the atmosphere and in the river or ocean.
- Apply manometer or other pressure instrument in a static fluid system.
- Calculate forces on submerged flat and curved surfaces.
- Understand Archimedes' principles.
- Calculate buoyancy on a submerged body, and on a floating body.
- Know pressure distribution in rigid-body motion.

Pressure

- A fluid at rest cannot support shear stress, i.e., **no shear stress in a static fluid**.
- The normal stress on any plane through a static fluid element (a point) is called the *fluid pressure*.
- Pressure is not a force, but it creates a force normal to the surface.
- Two important principles: (1) **no pressure change in horizontal direction**, (2) **a vertical change in pressure proportional to the density, gravity, and depth change**.

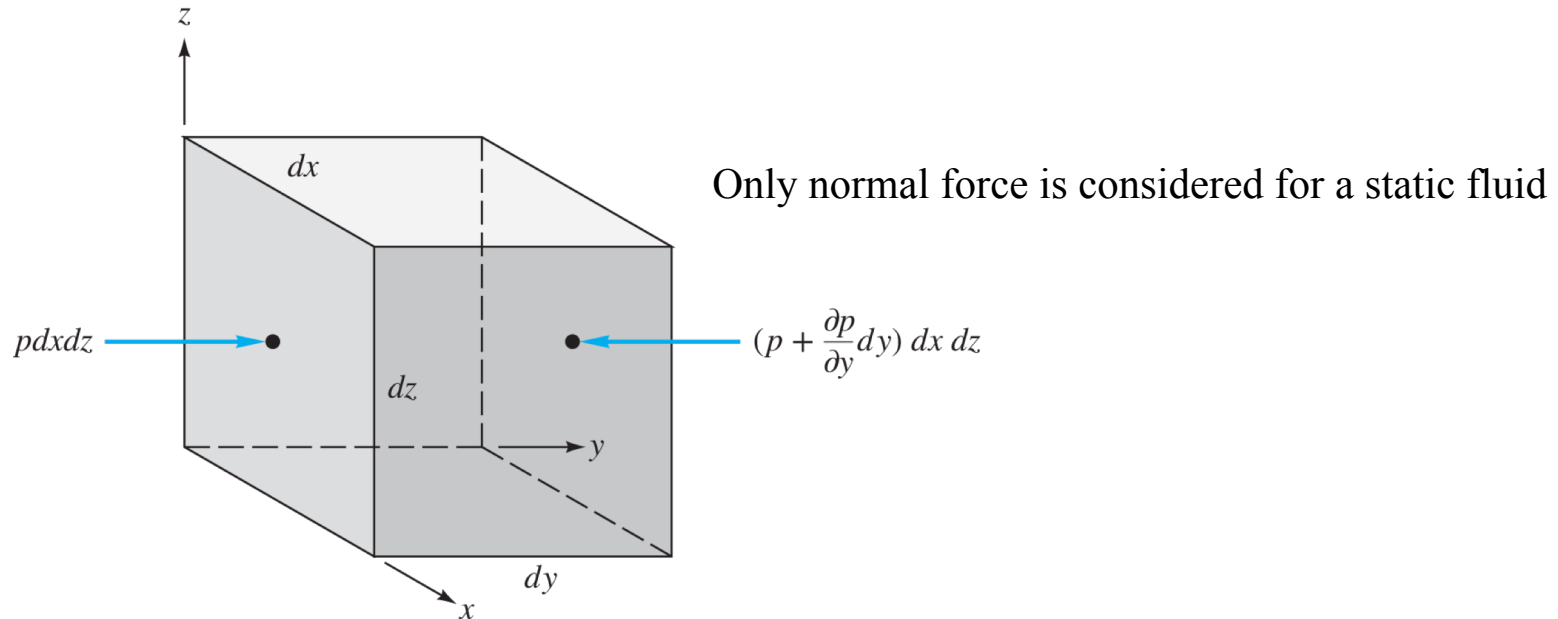


From force balance

$$p_x = p_n \quad p_z = p_n + \frac{1}{2} \rho g \Delta z$$

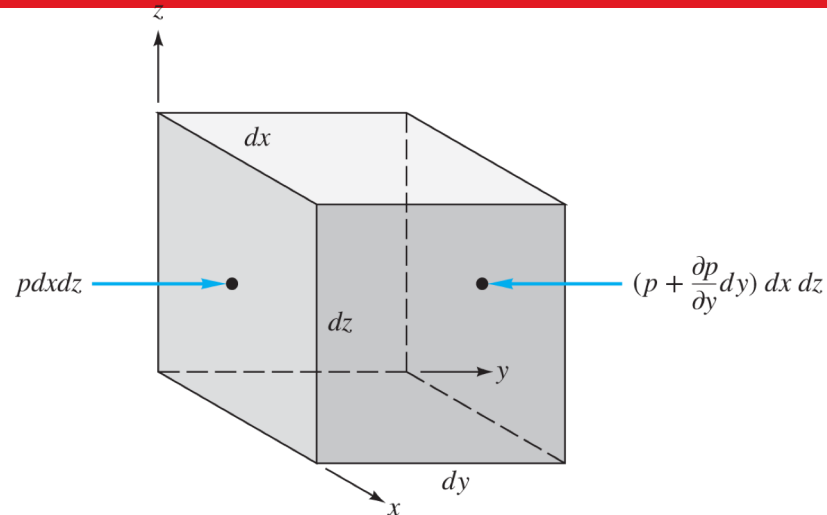
$$\text{If } \Delta z \rightarrow 0, \quad p_x = p_n = p_z$$

Equilibrium of a Fluid Element



- Pressure (or stress) causes a net force on a fluid element.
- It is not the pressure but the pressure gradient causing a net force.
- Pressure gradient is a surface force.

Equilibrium of a Fluid Element



The net force in x direction,

$$dF_x = p dx dz - \left(p + \frac{\partial p}{\partial x} dx \right) dy dz = -\frac{\partial p}{\partial x} dx dy dz$$

- The total net force vector on the element is:

$$d\mathbf{F}_{\text{press}} = \left(-\mathbf{i} \frac{\partial p}{\partial x} - \mathbf{j} \frac{\partial p}{\partial y} - \mathbf{k} \frac{\partial p}{\partial z} \right) dx dy dz \quad f_{\text{press}} = d\mathbf{F}_{\text{press}} / dx dy dz = -\nabla p$$

By Newton's law, the equilibrium of a force element is:

$$\sum \mathbf{f} = \mathbf{f}_{\text{press}} + \mathbf{f}_{\text{grav}} + \mathbf{f}_{\text{visc}} = -\nabla p + \rho \mathbf{g} + \mathbf{f}_{\text{visc}} = \rho \mathbf{a}$$

Equilibrium of a Fluid Element

- *Gage, absolute and vacuum pressures,*

Gauge Pressure

Pressure measured relative to atmospheric pressure (atmospheric pressure is taken as the zero reference). It represents how much a pressure is above or below atmospheric pressure.

$$p_{gauge} = p_{abs} - p_{atm}$$

Absolute Pressure

Pressure measured relative to a perfect vacuum (zero absolute pressure). It is the true, total pressure exerted at a point.

$$p_{abs} = p_{gauge} + p_{atm}$$

Vacuum Pressure

Used when the absolute pressure is *below* atmospheric pressure. It expresses the amount by which the pressure is below atmospheric pressure (i.e., a negative gauge pressure, expressed as a positive value).

$$p_{vacuum} = p_{atm} - p_{abs} \quad (\text{when } p_{abs} < p_{atm})$$

Hydrostatic Pressure Distribution

- If the fluid is at rest or at constant velocity, $\mathbf{a} = 0$ and $\mathbf{f}_{\text{visc}} = 0$.

$$\nabla p = \rho \mathbf{g} \rightarrow \frac{\partial p}{\partial x} = 0 \quad \frac{\partial p}{\partial y} = 0 \quad \frac{\partial p}{\partial z} = -\rho g = -\gamma \quad \text{or} \quad p_2 - p_1 = -\int_1^2 \gamma dz$$

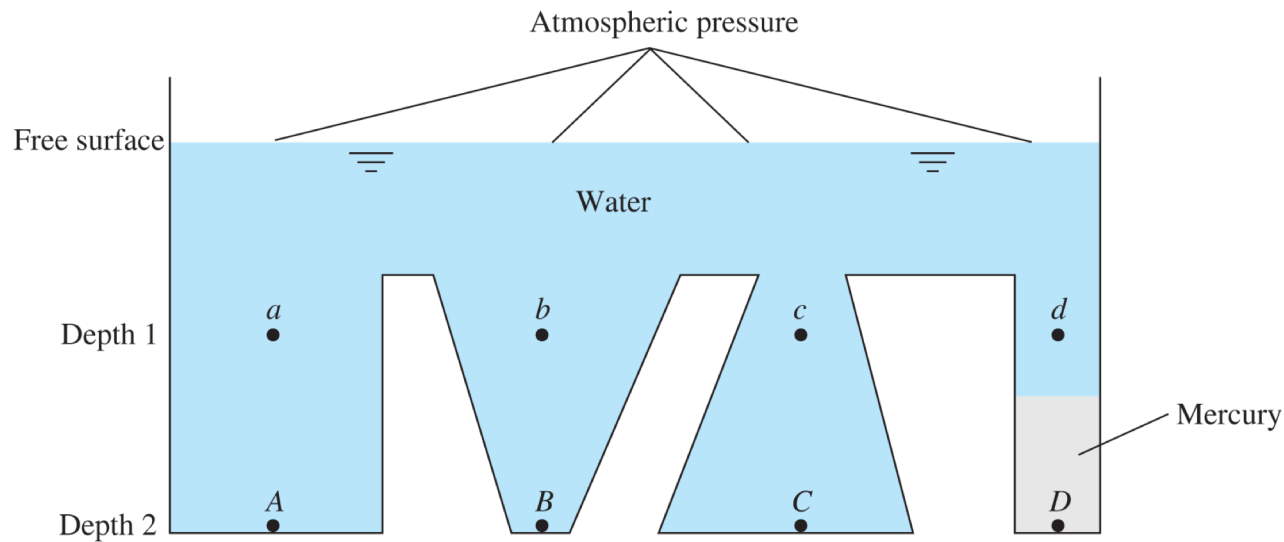
Specific weight

- Specific weight is especially useful for calculating **pressure and forces** in fluids. In hydrostatics, the pressure at depth h is:

$$p = \gamma h = \rho g h$$

Hydrostatic Pressure Distribution

- Pressure in a continuously distributed uniform static fluid varies only with vertical distance. The pressure is same at all points on a given horizontal plane in the fluid. The pressure increases with depth.



Hydrostatic pressure distribution

Hydrostatic Pressure in Liquids

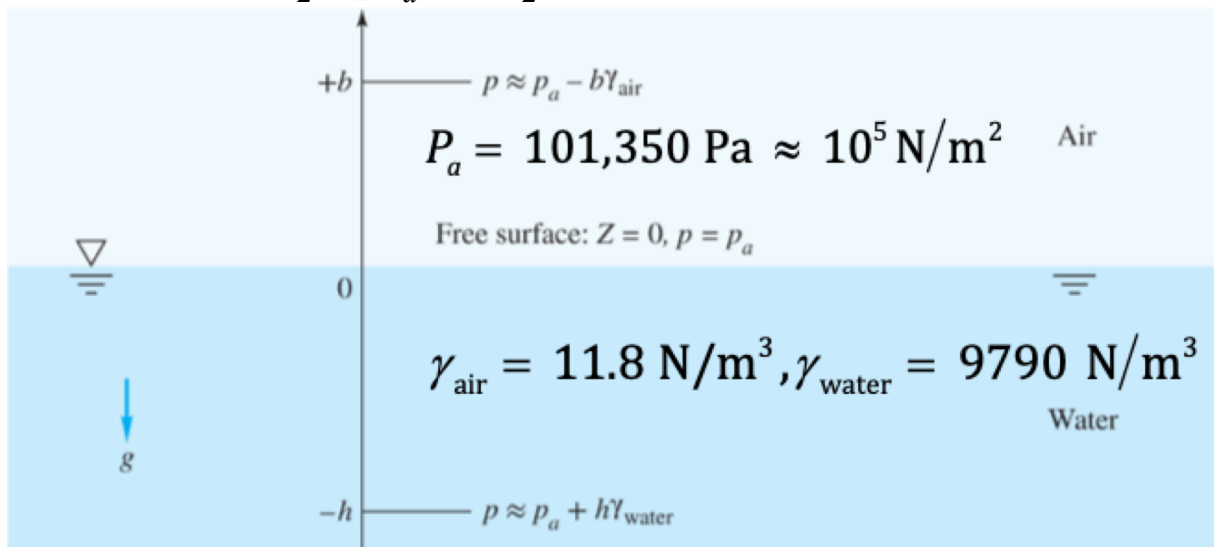
- Liquids are incompressible $\rightarrow$ constant density.

$$p_2 - p_1 = -\gamma(z_2 - z_1) \quad \text{or} \quad z_1 - z_2 = \frac{p_2}{\gamma} - \frac{p_1}{\gamma} \quad \leftarrow \text{Pressure head} \quad \left(h \text{ or } z = \frac{p}{\gamma} = \frac{p}{\rho g}\right)$$

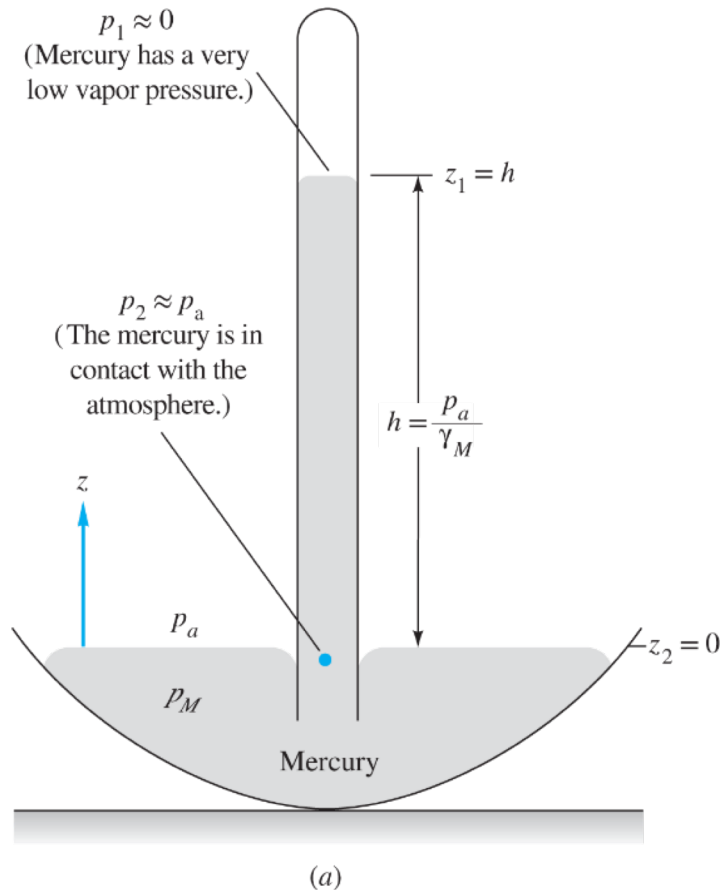
Pressure head represents the height of a column of a given fluid that would produce a certain pressure at its base due to its own weight.

Example: For rivers, lakes, and oceans, assume $z_1 = 0$, $p_1 = p_a$ at free surface.

Solution: $p_2 = p_a - \gamma z_2$



Hydrostatic Pressure in Liquids



Example: The mercury barometer, from the figure with $p_1 = 0$ (absolute) at $z_1 = h$ and $p_2 = p_a$ at $z_2 = 0$. What is h ?

Solution:

$$p_a - 0 = -\gamma_M (0 - h) \quad \text{or} \quad h = \frac{p_a}{\gamma_M}$$

At sea-level standard, $\gamma_M = 133,100 \text{ N/m}^3$ the barometric height is $h = 0.761 \text{ m}$ or 761 mm .

Hydrostatic Pressure in Gases

- Gases are compressible $\rightarrow$ ideal gas law: $p = \rho RT$.

$$\frac{dp}{dz} = -\rho g = -\frac{p}{RT} g \quad \text{or} \quad \int_1^2 \frac{dp}{p} = \ln \frac{p_2}{p_1} = -\frac{g}{R} \int_1^2 \frac{dz}{T}$$

- Isothermal atmosphere model, $T = T_0$, where T_0 is the sea level temperature. It is a rough approximation applied to low altitude.

$$p_2 = p_1 \exp \left[-\frac{g(z_2 - z_1)}{RT_0} \right]$$

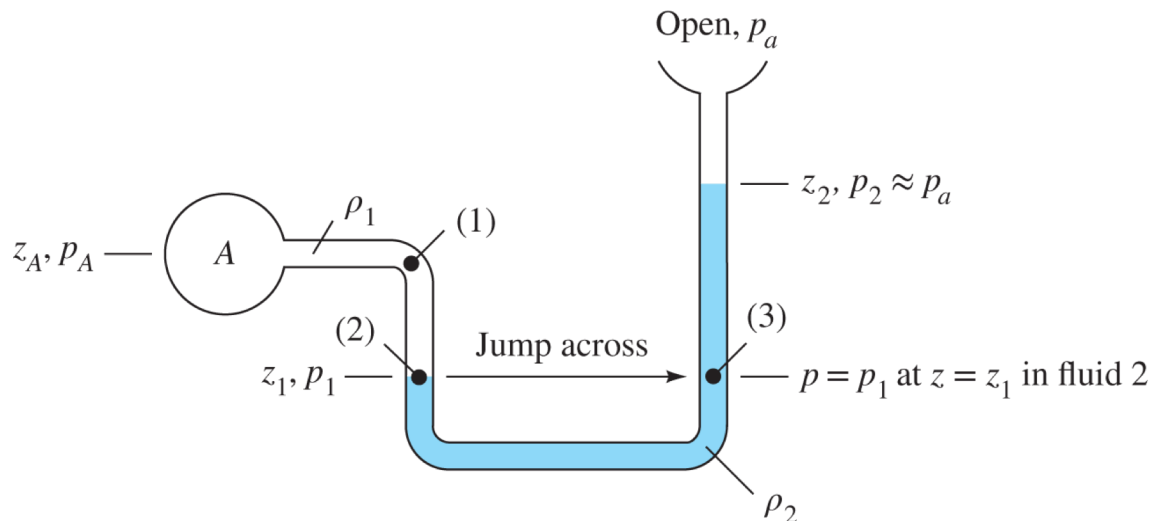
- Linear temperature drop model, $T \approx T_0 - Bz$, where B is the *lapse rate*. This is more accurate for the *troposphere* from 0 to 36,000 ft.

$$p = p_a \left(1 - \frac{Bz}{T_0} \right)^{g/RB} \quad \text{where} \quad \frac{g}{RB} = 5.26 \text{ (air)}$$

$$\rho = \rho_0 \left(1 - \frac{Bz}{T_0} \right)^{\frac{g}{RB} - 1} \quad \text{where} \quad \rho_0 = 1.2255 \text{ kg/m}^3$$

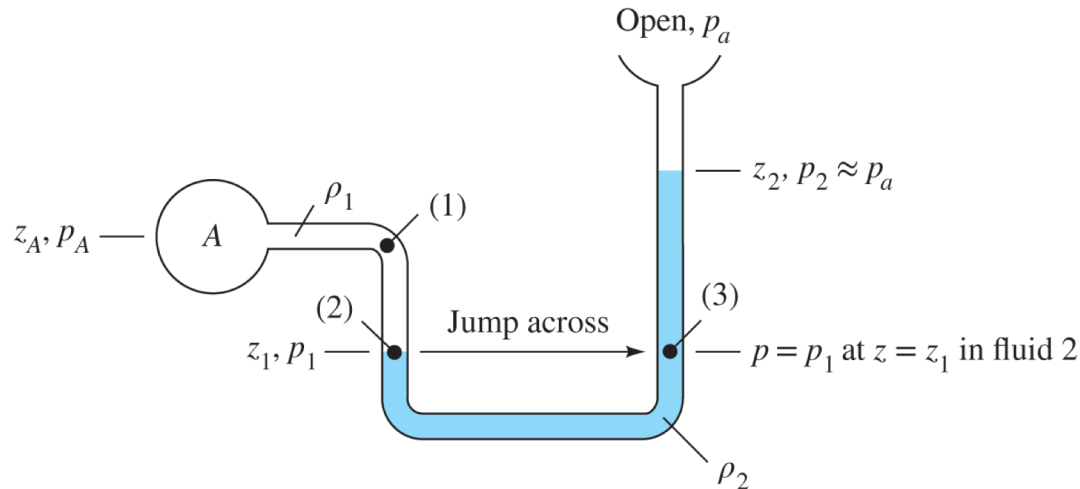
Application to Manometry

- The hydrostatic formula suggests a change in elevation $z_2 - z_1$ of a liquid is equivalent to a change in pressure $(p_2 - p_1)/\gamma$. Thus a static column of one or more liquids can be used to measure pressure differences. Such a device is called a manometer or U-tube.



Application to Manometry

Example: A simple manometer.



Solution 1:

$$p_A - p_1 = -\gamma_1 (z_A - z_1)$$

$$p_1 - p_2 = -\gamma_2 (z_1 - z_2)$$

$$p_A - p_2 = -\gamma_1 (z_A - z_1) - \gamma_2 (z_1 - z_2)$$

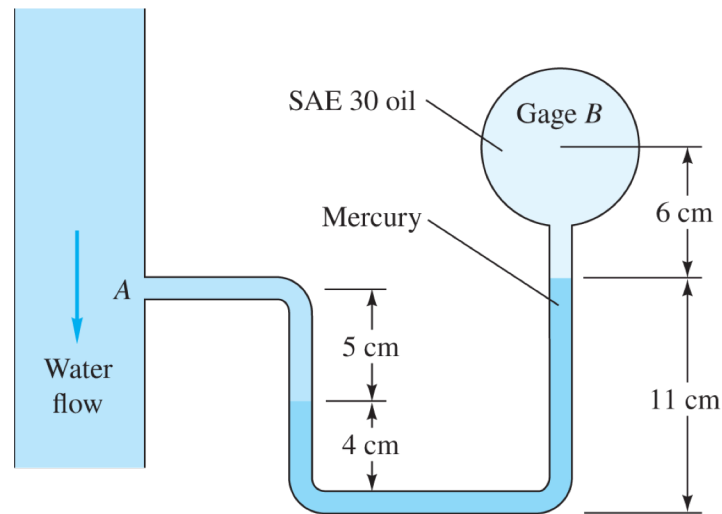
Solution 2:

$$p_A + \gamma_1 (z_A - z_1) - \gamma_2 (z_2 - z_1) = p_2$$

- **Pascal law:** Any two points at the same elevation in a continuous mass of the same static fluid will be at the same pressure – **Jump across**.

Application to Manometry

Example: Pressure gage B is to measure the pressure at A in a water flow. If the pressure at B is 87 kPa, estimate the pressure at A in kPa. Assume all fluids are at 20°C .



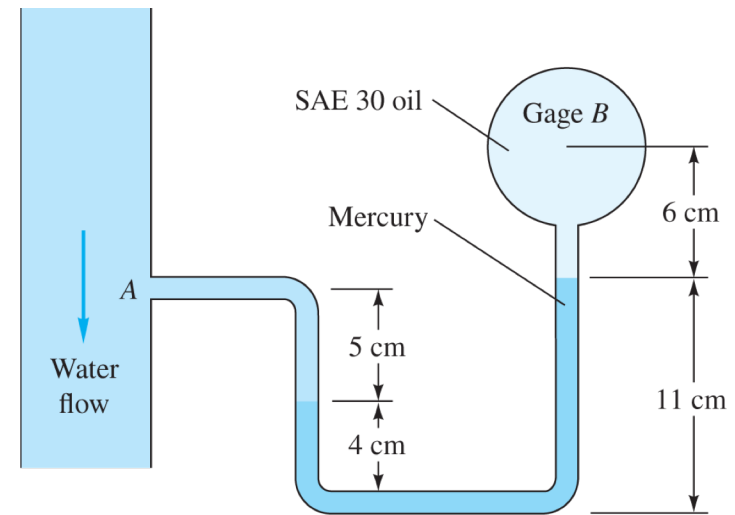
Application to Manometry

Solution:

From Table 2.1 or Table A.3: $\gamma_{\text{water}} = 9790 \text{ N/m}^3$, $\gamma_{\text{mercury}} = 133,100 \text{ N/m}^3$

$\gamma_{\text{oil}} = 8720 \text{ N/m}^3$

Proceed from A to B , “down” in water, jumping across at the left mercury meniscus, “up” in mercury, and “up” again in oil:



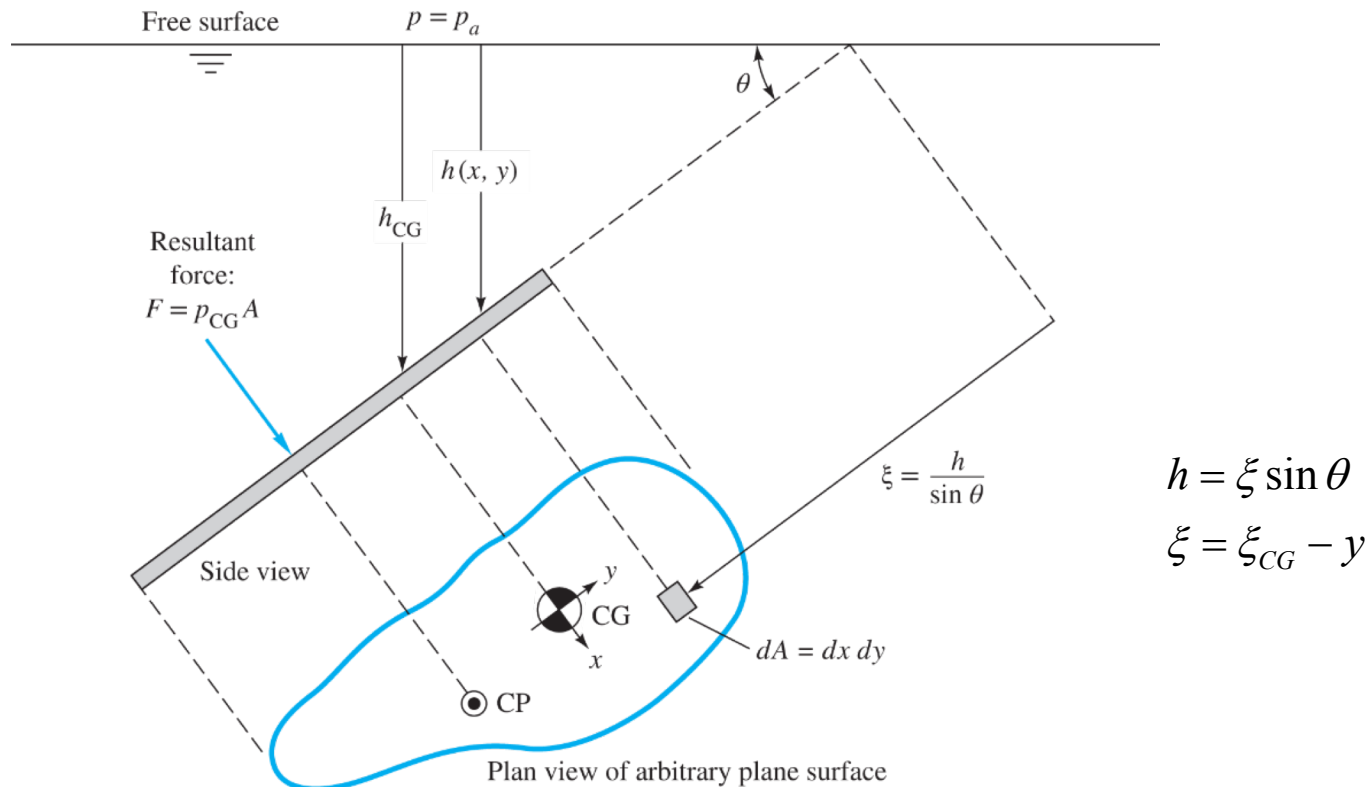
$$P_A + \gamma_w |\Delta z|_w - \gamma_m |\Delta z|_m - \gamma_o |\Delta z|_o = P_B$$

$$P_A + (9790 \text{ N/m}^3)(0.05 \text{ m}) - (133,100 \text{ N/m}^3)(0.07 \text{ m}) - (8720 \text{ N/m}^3)(0.06 \text{ m}) = 87,000$$

$$\text{Solve for } P_A = 96,350 \text{ N/m}^2 \approx 96.4 \text{ kPa}$$

Hydrostatic Forces on Plane Surfaces

- Hydrostatic forces relate to the pressure of fluid bearing on the surface.
- For a horizontal surface of area A at water depth h , $F = (p_a + \gamma h) A$
- For an inclined surface,



Hydrostatic Forces on Plane Surfaces

- For an inclined surface,

$$F = \int p dA \quad p = p_a + \gamma h$$

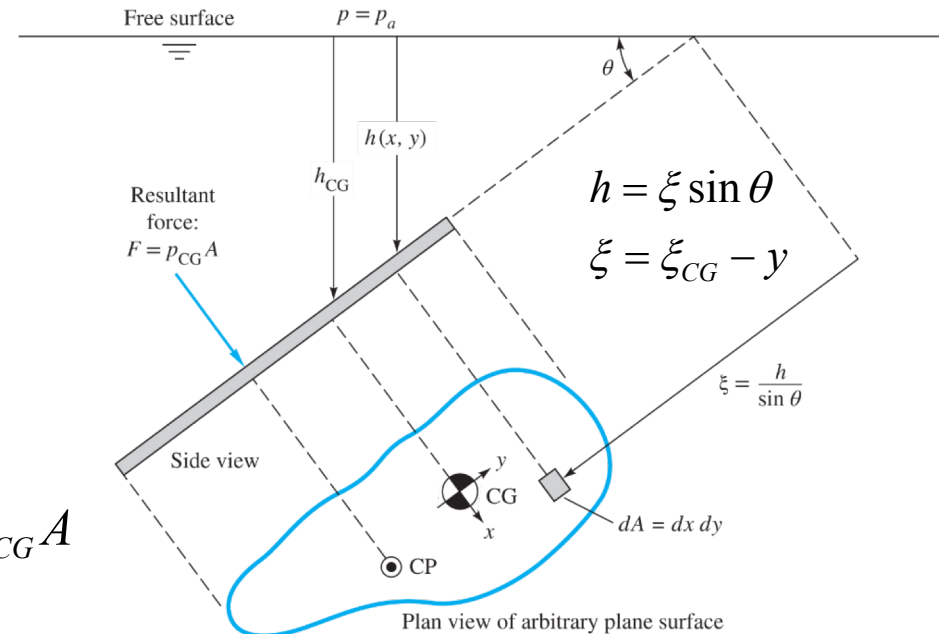
$$\text{So } F = \int (p_a + \gamma h) dA = p_a A + \gamma \int h dA$$

where $h = \xi \sin \theta$, by definition of centroid, the slant distance is:

$$\xi_{CG} = \frac{1}{A} \int \xi dA \quad \xi_{CG} \sin \theta = h_{CG}$$

$$F = p_a A + \gamma \sin \theta \int \xi dA = p_a A + \gamma \sin \theta \xi_{CG} A$$

$$\rightarrow F = p_a A + \gamma h_{CG} A = p_{CG} A$$

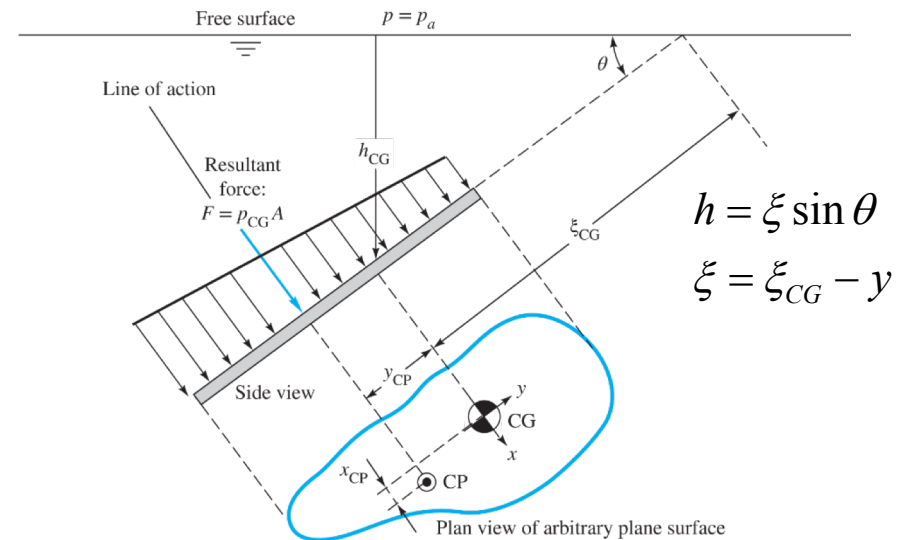


The magnitude of the force equals the pressure at the plate centroid times the plate area.

Where does the resultant force act at?

- The resultant force does not act through the centroid but below it toward the high-pressure side.
- **Center of pressure:** Resultant net force with **zero resultant moment** about that point
- To find the coordinates (x_{CP}, y_{CP}) , sum moments of the elemental force $p dA$ about the centroid,

$$F y_{CP} = \int y p dA = \gamma \sin \theta \int y \xi dA$$



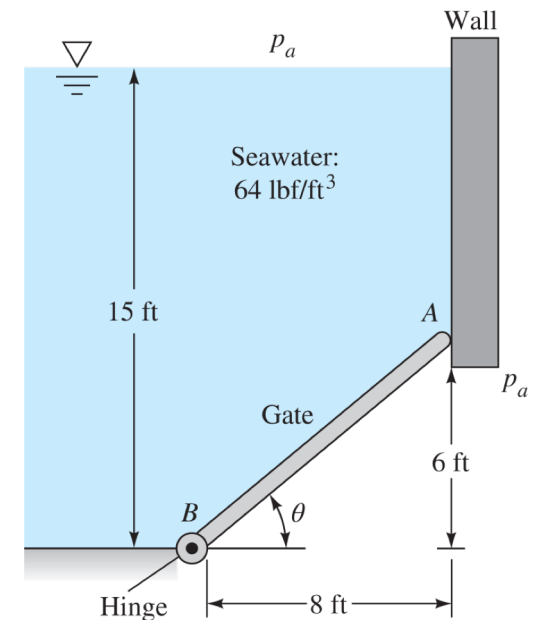
Hydrostatic Forces on Plane Surfaces

Example: The gate in figure is 5 ft wide, is hinged at point B , and rests against smooth wall at point A . Compute (a) the force on the gate due to seawater pressure, (b) the horizontal force P exerted by the wall at A , and (c) the reactions at the hinge B .

Solution:

(a) By geometry, the depth $h_{CG} = 15 - 3 = 12$ ft, the gate area $A = 5(10) = 50$ ft². Neglect p_a as acting on both sides of the gate,

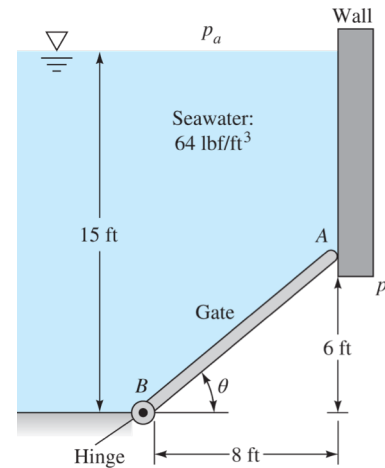
$$F = \gamma h_{CG} A = (64 \text{ lbf/ft}^3)(12 \text{ ft})(50 \text{ ft}^2) = 38,400 \text{ lbf}$$



Hydrostatic Forces on Plane Surfaces

(b) The gate is rectangular, hence

$$I_{xy} = 0 \quad \text{and} \quad I_{xx} = \frac{bL^3}{12} = \frac{(5 \text{ ft})(10 \text{ ft})^3}{12} = 417 \text{ ft}^4$$



To find the center of pressure force F , draw a FBD of the gate shown. The distance from the CG to the CP is:

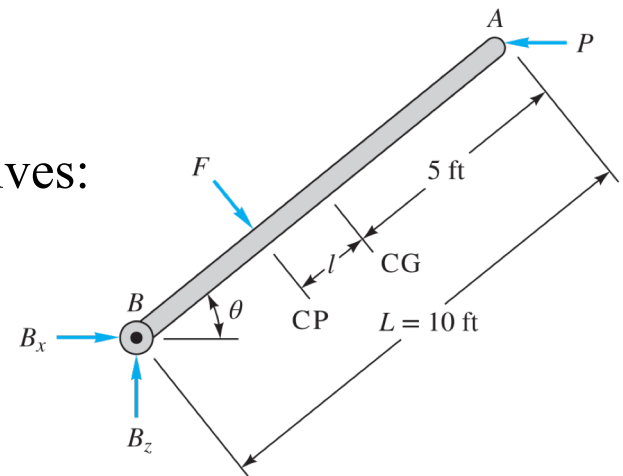
$$l = -y_{CP} = \frac{I_{xx} \sin \theta}{h_{CG} A} = \frac{(417 \text{ ft}^4)(6/10)}{(12 \text{ ft})(50 \text{ ft}^2)} = 0.417 \text{ ft}$$

Summing the moments counterclockwise about B gives:

$$PL \sin \theta - F(5 - l) = 0$$

$$P(6 \text{ ft}) - (38,400 \text{ lbf})(5 \text{ ft} - 0.417 \text{ ft}) = 0$$

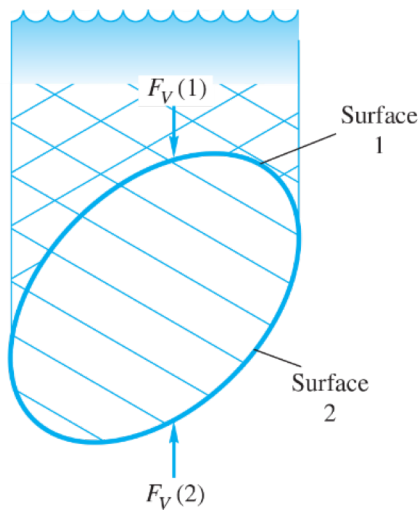
$$\text{So, } P = 23,900 \text{ lbf}$$



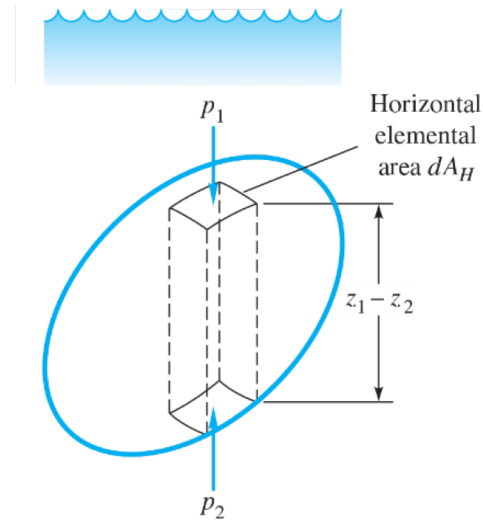
Buoyancy and Stability

- A body immersed in a fluid experiences a vertical buoyancy force equal to the weight of the fluid it displaces (Archimedes' law 1).

The center of buoyancy is the **centroid of the volume of liquid** displaced by the object.



$$\begin{aligned}
 F_B &= F_v(2) - F_v(1) \\
 &= (\text{fluid weight above 2}) - (\text{fluid weight above 1}) \\
 &= \text{weight of fluid equivalent to body volume}
 \end{aligned}$$



With uniform specific weight,

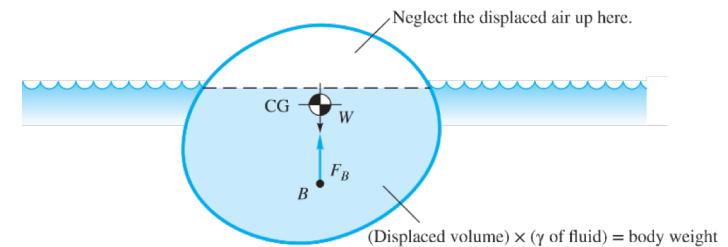
$$\begin{aligned}
 F_B &= \int (p_2 - p_1) dA_H \\
 &= -\gamma_f \int (z_2 - z_1) dA_H = \gamma_f V_B
 \end{aligned}$$

Buoyancy and Stability

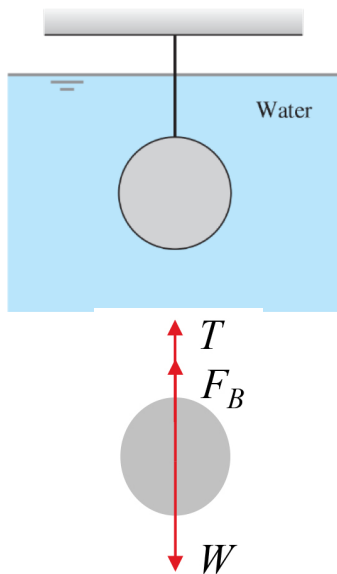
- A floating body displaces its own weight in the fluid: buoyancy force equals the body weight, and collinear (Archimedes' law 2).

$$F_B = \gamma_f V_{\text{displaced}} = W$$

A body will be neutrally buoyant if it has the right weight and volume for its ratio to equal the specific weight of the fluid.



Buoyancy Principle Applications



Example: A solid sphere in figure is iron ($SG = 7.9$). The tension in the table is 600 lbf. What is the diameter of the sphere in m?

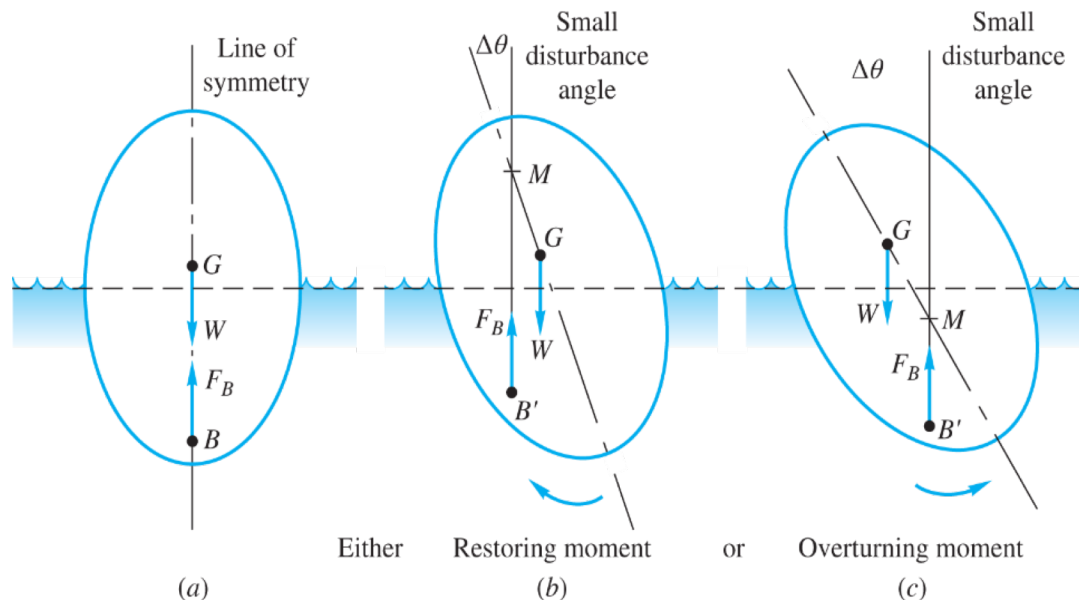
Solution: For water, $\gamma_w = 9790 \text{ N/m}^3$, the buoyancy force and sphere weight are $F_B = \gamma_w (\pi / 6) D^3$ and $W = 7.9 F_B$. By the force balance,

$$T = W - F_B = 7.9 F_B - F_B = 600 \text{ lbf} = 2670 \text{ N}$$

$$6.9 \gamma_w (\pi / 6) D^3 = 2670 \text{ or } D = 0.42 \text{ m}$$

Stability

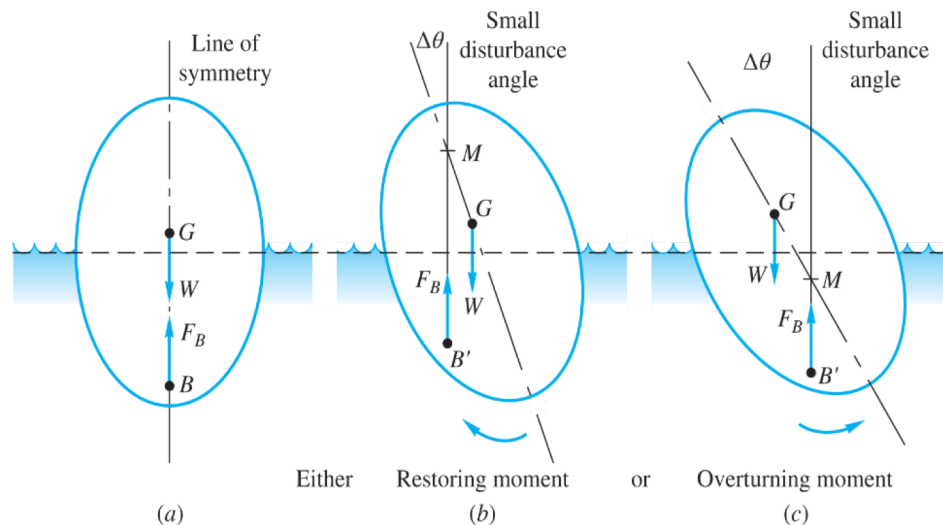
- A floating body may not be statically stable. If it will turn at first opportunity, and it is said to be *statically unstable*.
- **The only way to tell whether a floating position is stable is to “disturb” the body a slight amount and see whether it develops a restoring moment which will return it to its original position.**
- The center of buoyancy is the **centroid of the volume** of liquid displaced by the object. It moves when a floating object tilts or heels over, because the shape of the underwater volume changes.



Stability

Steps for the static stability design for a symmetric floating body:

- Calculating F_B , position of mass center G , and center of buoyancy B are needed.



- Tilting an small angle $\Delta\theta$, the new center of buoyancy B' is calculated, a vertical line drawn up intersects the line of symmetry at a point M , called metacenter, which is independent of $\Delta\theta$ for small angle.
- If M is above G , metacenter height $\overline{MG} > 0$ restoring moment is present, the original position stable. If M is below G , $\overline{MG} < 0$, the body is unstable. Stability increases with increasing $\overline{MG}$.

Pressure Distribution in Rigid-Body Motion

- Uniform linear acceleration: acceleration $\mathbf{a}$ having the same magnitude and direction for all particles.
- With no relative motion, there are no viscous force $\mu \nabla^2 \mathbf{V} = 0$
- The equilibrium of a force element comes from pressure, gravity, and acceleration.

$$\nabla p = \rho(\mathbf{g} - \mathbf{a})$$

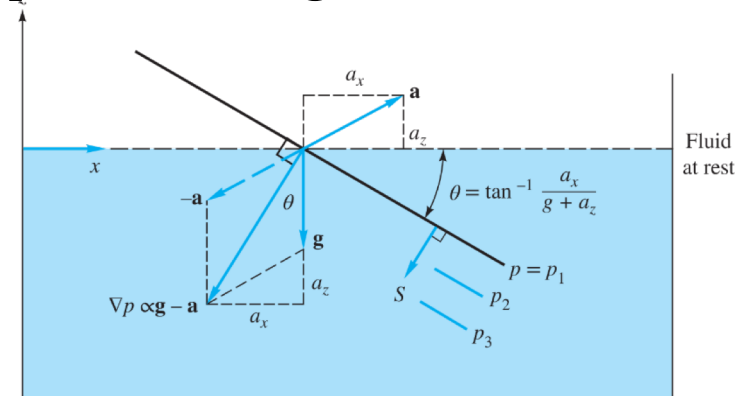
- The surfaces of constant pressure are perpendicular to $\mathbf{g} - \mathbf{a}$, and are tilted as:

$$\theta = \tan^{-1} \frac{a_x}{g + a_z}$$

The rate of increase of pressure in the direction of $\mathbf{g} - \mathbf{a}$

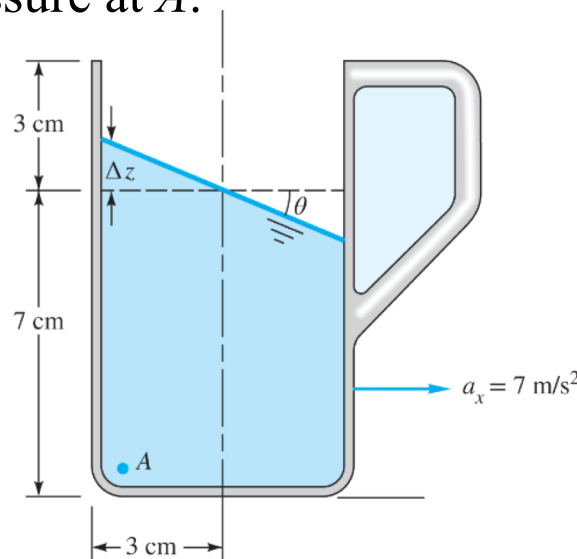
$$\frac{dp}{ds} = \rho G$$

$$\text{where } G = \left[a_x^2 + (g + a_z)^2 \right]^{1/2}$$



Pressure Distribution in Rigid-Body Motion

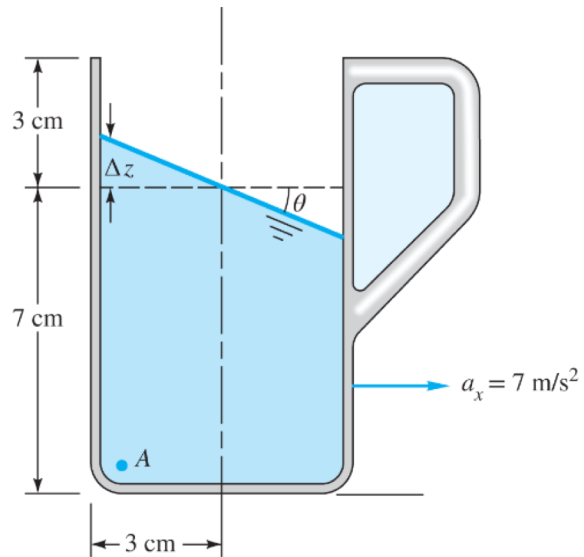
Example: The coffee in the mug tilted during the horizontal acceleration, assuming symmetric mug. Calculate (a) the angle of tilt, (b) height of coffee rise, and (3) the gauge pressure at A .



Solution: (a) The angle of tilt is $\theta = \tan^{-1} \frac{a_x}{g} = \tan^{-1} \frac{7.0 \text{ m/s}^2}{9.81 \text{ m/s}^2} = 35.5^\circ$

(b) The real side coffee rise is $\Delta z = (3 \text{ cm}) \tan 35.5^\circ = 2.14 \text{ cm}$

Pressure Distribution in Rigid-Body Motion



(c) The gauge pressure at A is

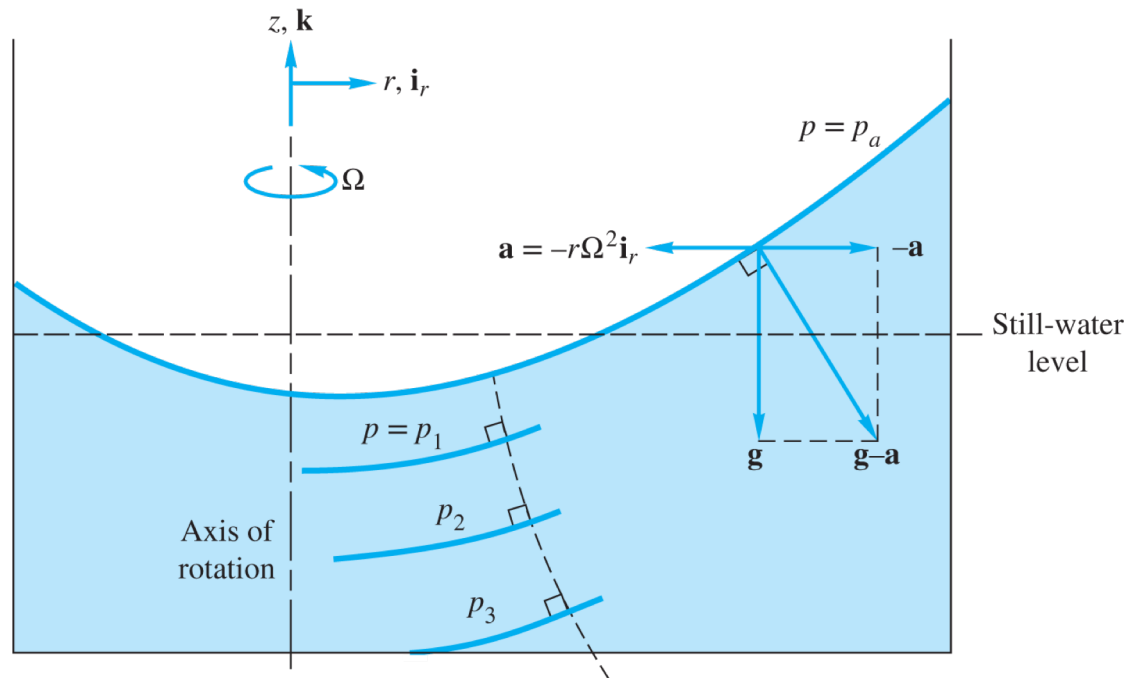
- *Approach (b):* The pressure at A can be computed from Eq. (2.40), using the perpendicular distance Δs from the surface to A . When at rest, $p_A = \rho g h_{\text{rest}} = (1010 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(0.07 \text{ m}) = 694 \text{ Pa}$.

$$p_A = \rho G \Delta s = \left(1010 \frac{\text{kg}}{\text{m}^3}\right) \left[\sqrt{(9.81)^2 + (7.0)^2} \right] [(0.07 + 0.0214) \cos 35.5^\circ] \approx 906 \text{ Pa} \quad \text{Ans. (b)}$$

$$p_A = \rho g (z_{\text{surf}} - z_A) = (1010 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(0.0214 \text{ m} + 0.07 \text{ m}) = 906 \text{ Pa}$$

Rigid-Body Rotation

- Assume the container has been rotating long enough at constant Ω for the fluid to have attained rigid-body rotation. $\mathbf{\Omega} = \mathbf{k}\Omega$ $\mathbf{r}_0 = \mathbf{i}_r r$
- Acceleration is a centripetal term. $\mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{r}_0) = -r\Omega^2 \mathbf{i}_r$



Rigid-Body Rotation

- The force balance becomes,

$$\nabla p = \mathbf{i}_r \frac{\partial p}{\partial r} + \mathbf{k} \frac{\partial p}{\partial z} = \rho(\mathbf{g} - \mathbf{a}) = \rho(-g\mathbf{k} + r\Omega^2\mathbf{i}_r)$$

$$\frac{\partial p}{\partial r} = \rho r \Omega^2 \quad \frac{\partial p}{\partial z} = -\gamma$$

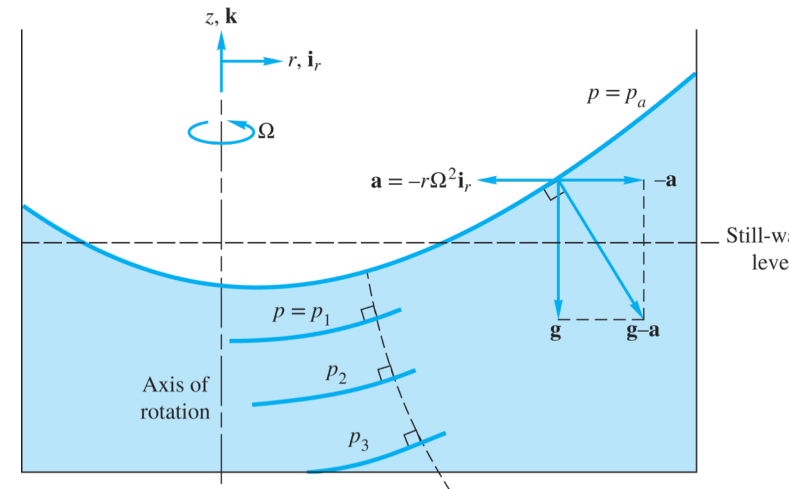
Integrate and determine the constant:

$$p = p_0 - \gamma z + \frac{1}{2} \rho r^2 \Omega^2$$

The pressure is linear in z and parabolic in r .

$$\rightarrow z = \frac{p_0 - p_1}{\gamma} + \frac{r^2 \Omega^2}{2g} = a + br^2$$

- The surfaces are paraboloids of revolution, concave upward, with their minimum on the axis of rotation.



Summary

- Pressure (or stress) causes a net force on a fluid element.

$$\sum \mathbf{f} = \mathbf{f}_{\text{press}} + \mathbf{f}_{\text{grav}} + \mathbf{f}_{\text{visc}} = -\nabla p + \rho \mathbf{g} + \mathbf{f}_{\text{visc}} = \rho \mathbf{a}$$

- If the fluid is at rest or at constant velocity, $\mathbf{a} = 0$ and $\mathbf{f}_{\text{visc}} = 0$.

$$\nabla p = \rho \mathbf{g} \rightarrow \frac{\partial p}{\partial x} = 0 \quad \frac{\partial p}{\partial y} = 0 \quad \frac{\partial p}{\partial z} = -\rho g = -\gamma$$

- Manometer measures pressure difference based on hydrostatic pressure distribution in liquids. $p_2 - p_1 = -\gamma(z_2 - z_1)$
- The magnitude of the hydrostatic force on a plane surface equals the pressure at the plate centroid times the plate area.

$$F = \gamma h_{CG} A \quad y_{CP} = -\gamma \sin \theta \frac{I_{xx}}{p_{CG} A} \quad x_{CP} = -\gamma \sin \theta \frac{I_{xy}}{p_{CG} A}$$

- A body immersed in a fluid experiences a vertical buoyancy force. A floating body displaces its own weight in the fluid.
- For a rigid-body motion of fluid, the pressure distribution is $\nabla p = \rho(\mathbf{g} - \mathbf{a})$. Special cases of linear acceleration and fixed axis rotation are discussed.